Systems of Ordinary Differential Equations

1. Models of Pursuit
   - the tractrix problem
   - a predator chasing a prey
2. Multiple Mass Springs and Traffic
   - multiple mass spring systems
   - a car following theory
3. Nonlinear Systems
   - administering half a pill a day
   - an arms race

MCS 472 Lecture 31
Industrial Math & Computation
Jan Verschelde, 1 April 2026

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the tractrix problem

Consider a trailer attached to a tractor.

For example: a toddler pulling the string attached to a toy car.

Input: $(x_1(t), x_2(t))$ is the path of the tractor.

Output: $(y_1(t), y_2(t))$ is the path of the trailer.

The coordinate functions $y_1(t)$ and $y_2(t)$ are the solutions of a *system* of two ordinary differential equations.

deriving the differential equations

Input: $(x_1(t), x_2(t))$ is the path of the tractor.

Output: $(y_1(t), y_2(t))$ is the path of the trailer.

The velocity vector of the trailer is parallel
to the direction of the bar that connects trailer to tractor:

$$\left(\begin{array}{c} y_1' \\ y_2' \end{array} \right) = \lambda \left(\begin{array}{c} x_1 - y_1 \\ x_2 - y_2 \end{array} \right), \quad \lambda > 0.$$

- $(y_1'(t), y_2'(t))$ is the change in position of the trailer.
- $(x_1(t) - y_1(t), x_2(t) - y_1(t))$ is the direction vector.

projecting the direction vector

The velocity vector of the trailer is the projection of the direction vector of the trailer onto the direction of the bar:

$$u = \frac{(x_1 - y_1, x_2 - y_2)}{||(x_1 - y_1, x_2 - y_2)||} \quad \text{and} \quad v = (x_1', x_2'),$$

we compute the projection of the velocity vector as $(v^T u)u$.

Then the system of first-order equations that defines the path of the trailer is given by

$$\begin{pmatrix} y_1' \\ y_2' \end{pmatrix} = (v^T u)u,$$

where

- u is the normalized vector of the direction of the bar, and
- v is the velocity vector of the tractor.

the tractor moves on a circle, counter clockwise

```
"""
    function tractor(time::Float64)

returns a 4-tuple xp, yp, xv, yv, with
the coordinates xp, yp of the position, and
the components xv, yv of the velocity vector
for the tractor at the given time.
"""
function tractor(time::Float64)
    xpos = cos(time)
    ypos = sin(time)
    xvel = -sin(time)
    yvel = cos(time)
    return (xpos, ypos, xvel, yvel)
end
```

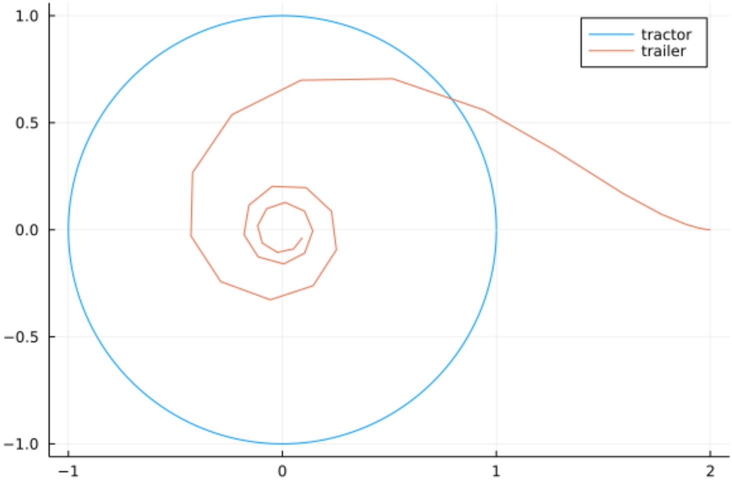
the right hand side of the system

```
"""
    function trailer!(du::Vector{Float64},
                      u::Vector{Float64}, p, t)

defines the right hand side of the system.
"""
function trailer!(du::Vector{Float64},
                  u::Vector{Float64}, p, t)
    x1p, x2p, x1v, x2v = tractor(t)
    du[1] = x1p - u[1]
    du[2] = x2p - u[2]
    nrm = sqrt(du[1]*du[1] + du[2]*du[2])
    du[1] = du[1]/nrm
    du[2] = du[2]/nrm
    vtu = x1v*du[1] + x2v*du[2]
    du[1] = vtu*du[1]
    du[2] = vtu*du[2]
end
```

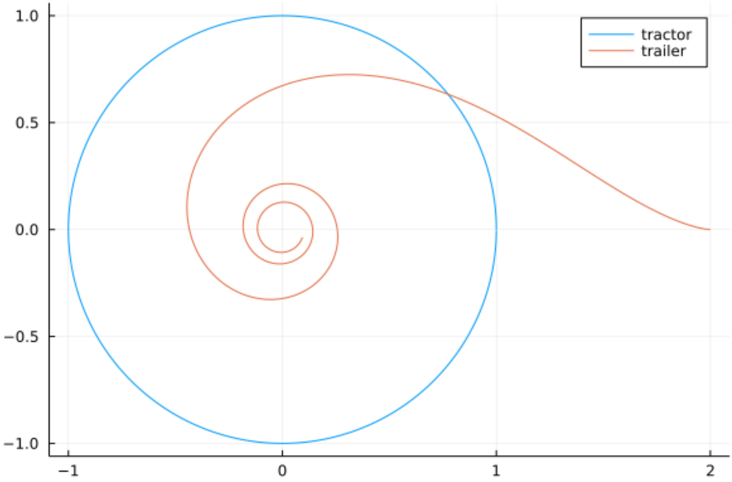
solving and plotting

```
timespan = (0, 20)

initconds = [2.0, 0.0]

odetractrix = ODEProblem(trailer!, initconds, timespan)

soltractrix = solve(odetractrix)

trailerpath = [(x,y) for (x,y) in soltractrix.u]

runcircle = [tractor(t) for t=0:0.01:2*pi]
tractorpath = [(tpl[1],tpl[2]) for tpl in runcircle]

plot(tractorpath, label="tractor")
plot!(trailerpath, label="trailer")
```

solution trajectory



solution trajectory with `saveat=0.05`



defining an animation

```
animTractorTrailer = Plots.Animation()
for i = 10:10:length(tractorpath2)
    plot(tractorpath2[1:i], label="tractor",
         xlims=(-1.1,2.1), ylims = (-1.1, 1.1))
    (xtp1, xtp2) = tractorpath2[i]
    scatter!([xtp1], [xtp2], legend=false)
    plot!(trailerpath2[1:i], label="trailer")
    Plots.frame(animTractorTrailer)
end

gif(animTractorTrailer, "tractortrailer2.gif",
    fps = 2)
```

See the posted Jupyter notebook.

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a predator chasing a prey

Input: $(x_1(t), x_2(t))$ is the path of the prey.

Output: $(y_1(t), y_2(t))$ is the path of the predator.

The velocity vector of the predator is parallel
to the direction of predator towards the prey:

$$\left(\begin{array}{c} y_1' \\ y_2' \end{array} \right) = \lambda \left(\begin{array}{c} x_1 - y_1 \\ x_2 - y_2 \end{array} \right), \quad \lambda > 0.$$

- $(y_1'(t), y_2'(t))$ is the change in position of the trailer.
- $(x_1(t) - y_1(t), x_2(t) - y_1(t))$ is the direction vector.

the adjusted tractrix model

The velocity vector of the predator is the projection
of the direction vector of the predator towards the prey:

$$\left(\begin{array}{c} y_1' \\ y_2' \end{array} \right) = s(t)\, u,$$

with

$$u = \frac{(y_1 - x_1, y_2 - x_2)}{||(y_1 - x_1, y_2 - x_2)||}$$

where $s(t)$ is the speed of the predator,
for example:

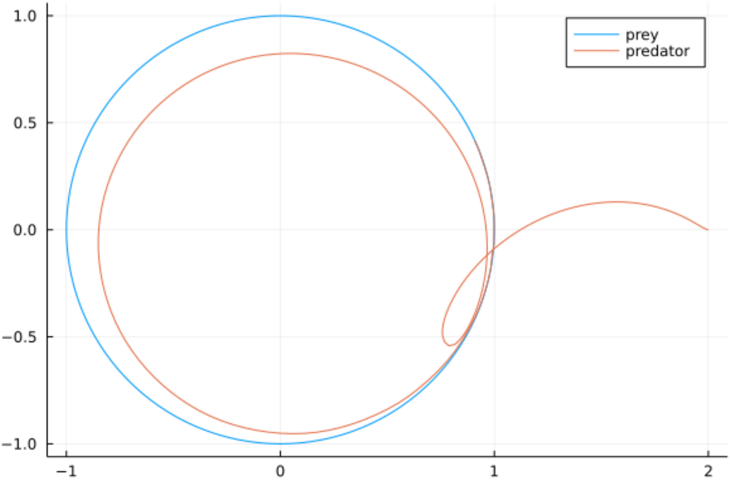
- $s(t) = c$, at constant speed c, or
- $s(t) = a\,t$, with acceleration a.

code for the predator

```
"""
    function predator!(du::Vector{Float64},
                       u::Vector{Float64}, p, t)

defines the right hand side of the system for the
predator, chasing a prey with accelerating speed.
"""
function predator!(du::Vector{Float64},
                   u::Vector{Float64}, p, t)
    x1p, x2p, x1v, x2v = tractor(t)
    acceleration = p
    speed = acceleration*t
    du[1] = x1p - u[1]
    du[2] = x2p - u[2]
    nrm = sqrt(du[1]*du[1] + du[2]*du[2])
    du[1] = speed*du[1]/nrm
    du[2] = speed*du[2]/nrm
end
```

a predator in pursuit of a prey, acceleration = 0.1



interpretation of the trajectory

Look at the animation in the posted Jupyter notebook.

The speed of the predator is $s(t) = 0.1t$.

- For small t, the speed $0.1t$ is very small, and the predator moves very slowly.
- Once t goes past 10, the speed of the predator becomes larger than one.
- The prey remains circling at unit speed.
- So eventually, the predator catches the prey.

rabbits and foxes

- Let $r(t)$ denote the population of the rabbits.
- Let $f(t)$ denote the population of the foxes.

The system that models their evolution is

$$\left\{ \begin{array}{rcl} \frac{d}{dt} r(t) & = & a r(t) - b r(t) f(t), \\ \frac{d}{dt} f(t) & = & -c f(t) + d r(t) f(t). \end{array} \right.$$

Exercise 1:

1. Let $a = 1.1$, $b = 0.5$, $c = 0.75$, $d = 0.25$, $r(0) = 3$, and $f(0) = 2$. Set up the system of differential equations and solve it, for the time span $[0, 10]$.
2. Plot the trajectories for $r(t)$ and $f(t)$. What is the end value for t for one complete orbit?

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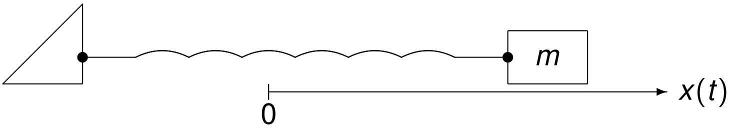
one spring mass system

Consider a mass m sliding along a frictionless horizontal table, restrained by a spring. The displacement from equilibrium is $x(t)$.



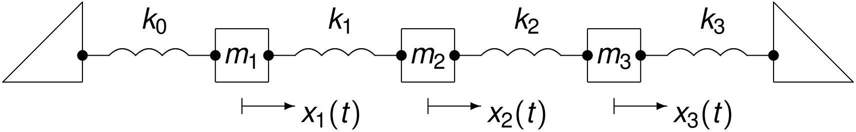
By Newton's law, the inertial force is $m\frac{d^2}{dt}x(t)$.

The restoring force of the spring is proportional to $x(t)$, so we obtain

$$m\frac{d^2}{dt}x(t) + kx(t) = 0, \quad \text{for some constants } k.$$

multiple mass spring system

Three masses m_1, m_2, m_3, connected by springs:



The second mass receives a push

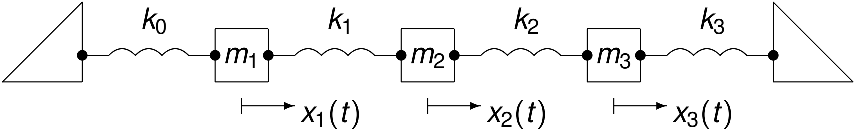
- from m_1 with spring constant k_1, and
- from m_3 with spring constant k_2,

compensated by the sum of k_1 and k_2:

$$m_2 \frac{d^2}{d\,t^2} x_2(t) = k_1 x_1(t) - (k_1 + k_2) x_2 + k_2 x_3(t).$$

multiple mass spring system

Three masses m_1, m_2, m_3, connected by springs:



$$
\begin{aligned}
m_1 x_1''(t) &= - (k_0 + k_1)x_1 + k_1 x_2(t) \\
m_1 x_2''(t) &= k_1 x_1(t) - (k_1 + k_2)x_2 + k_2 x_3(t) \\
m_1 x_3''(t) &= k_2 x_2(t) - (k_2 + k_3)x_3
\end{aligned}
$$

Exercise 2:
Write the above system of second order differential equations as a system of first order differential equations, in matrix format.

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a car following theory

Consider a platoon of cars:

- $x_1(t)$ is the position of the first vehicle at time t,
- $x_k(t)$ is the position of the kth vehicle in the platoon.

What happens in reaction to a disturbance?

A model for the change in velocity caused by a difference in the velocity of the previous vehicle is

$$x_k''(t + \Delta t) = \alpha \left(x_{k-1}'(t) - x_k'(t) \right), \quad \text{for } k > 1,$$

where

- Δt is the reaction time of the kth driver,
- α is a sensitivity parameter.

Empirical values: $\Delta t \approx 1.55$ sec and $\alpha \approx 0.37$ sec^{-1}.

the distances between the vehicles

Distance equals speed multiplied with the time traveled.

$$\begin{aligned} d_k(t) &= x_{k-1}(t) - x_k(t) \\ &= \int_0^t \left(x'_{k-1}(s) - x'_k(s) \right) ds + d, \end{aligned}$$

where d is the initial distance between the vehicles. Use

$$x''_k(t + \Delta t) = \alpha \left(x'_{k-1}(t) - x'_k(t) \right)$$

to obtain

$$\begin{aligned} d_k(t) &= \int_0^t \frac{x''_k(s + \Delta t)}{\alpha} ds + d \\ &= \frac{1}{\alpha} \left(v_k(t + \Delta t) - v_k(0 + \Delta t) \right) + d. \end{aligned}$$

a condition to avoid collision

The distance between two vehicles:

$$\begin{aligned} d_k(t) &= \frac{1}{\alpha}\Big(v_k(t+\Delta t) - v_k(0+\Delta t) \Big) + d \\ &= \frac{1}{\alpha} v_k(t+\Delta t) - \frac{v}{\alpha} + d, \end{aligned}$$

where $v \approx v(\Delta t)$, when Δt is small enough.

Deriving an upper limit on the speed to avoid a collision:

$$-\frac{v}{\alpha} + d > 0 \quad \Rightarrow \quad v < \alpha d \quad \text{or} \quad d > v/\alpha.$$

- $v < \alpha d$ is an upper limit on the speed, or
- $d > v/\alpha$ is a lower bound on the distance.

$\alpha \approx 0.37$ sec^{-1}, $1/\alpha \approx 2.7$ sec, which gives rise to the 3 second rule.

the three second rule

We derived $d > v/\alpha$ as a lower bound on the distance to avoid a collision.

The empirical values

$$\alpha \approx 0.37 \text{ sec}^{-1} \quad \text{and} \quad 1/\alpha \approx 2.7 \text{ sec},$$

give rise to the 3 second rule, as $d > 2.7v$.

The distance between two vehicles should be larger than the distance traveled in three seconds.

For information on Defensive Driving Courses, see the publication of the National Safety Council, 2019, on "*The DDC Instructor And Administrative Reference Guide.*"

expressway traffic flow

Proposal for a Topic of a Project:

Take data on expressway traffic.

At various times of the day:

- What is the separation between cars?
- What is the separation versus velocity?
- How is flow related to density?

When is the road most dangerous?

systems of ordinary differential equations

administering half a pill a day

We administer half a pill from a bottle, each day.

In taking a pill from the bottle:

- If it is a half pill, then we just take it.
- If the pill is whole, then
  - break the pill in half, keep one half,
  - throw the other half in the bottle.

Here is then our question:

After k days, how many whole pills are left in the bottle?

the equation on the number of whole pills

- Let x_k the expected number of whole pills, at day k.
- Let y_k the expected number of half pills, at day k.

Consider now day $k + 1$:

$$x_{k+1} - x_k = -\frac{x_k}{x_k + y_k}$$

is the expected net change in whole pills,
as the number will decrease only if whole pill is drawn.

The ratio $\dfrac{x_k}{x_k + y_k}$ is the ratio

- of the number of whole pills,
- over the total number of pills.

the equation on the number of half pills

For half pills:

$$y_{k+1} - y_k = \frac{x_k}{x_k + y_k} - \frac{y_k}{x_k + y_k},$$

which is

- the increase if a whole pill is drawn, and
- the decrease if a half pill is drawn.

Next, we introduce expected percentages

$$p_k = \frac{x_k}{N} \quad \text{and} \quad q_k = \frac{y_k}{N}.$$

rewriting with expected percentages

The equations on the expected number of whole and half pills

$$
\begin{aligned}
x_{k+1} - x_k &= -\frac{x_k}{x_k + y_k} \\
y_{k+1} - y_k &= \frac{x_k}{x_k + y_k} - \frac{y_k}{x_k + y_k}
\end{aligned}
$$

is then rewritten with $p_k = x_k/N$ and $q_k = y_k/N$ into

$$
\begin{aligned}
\frac{p_{k+1} - p_k}{1/N} &= -\frac{p_k}{p_k + q_k} \\
\frac{q_{k+1} - q_k}{1/N} &= \frac{p_k}{p_k + q_k} - \frac{q_k}{p_k + q_k}.
\end{aligned}
$$

a nonlinear system

The original discrete formulation

$$
\begin{aligned}
\frac{p_{k+1} - p_k}{1/N} &= -\frac{p_k}{p_k + q_k} \\
\frac{q_{k+1} - q_k}{1/N} &= \frac{p_k}{p_k + q_k} - \frac{q_k}{p_k + q_k}
\end{aligned}
$$

has a continuous version:

$$
\begin{aligned}
\frac{d}{dt} p(t) &= -\frac{p}{p+q} \\
\frac{d}{dt} q(t) &= \frac{p}{p+q} - \frac{q}{p+q}
\end{aligned}
$$

with initial conditions $p(0) = 1$ and $q(0) = 0$.

Note that the system is deterministic.

solve the system

Exercise 3: Consider

$$\begin{aligned}\frac{d}{dt}p(t) &= -\frac{p}{p+q}\\ \frac{d}{dt}q(t) &= \frac{p}{p+q} - \frac{q}{p+q}\end{aligned}$$

with initial conditions $p(0) = 1$ and $q(0) = 0$.

- Solve the system.
- Interpret the solution.

systems of ordinary differential equations

an arms race

Two nations X and Y are competing in an arms race.

At year n,

$$x_n = \frac{\text{arms expenditure of nation } X}{\text{gross national product of } X}$$

$$y_n = \frac{\text{arms expenditure of nation } Y}{\text{gross national product of } Y}$$

We have $0 < x_n, y_n < 1$.

Both nations are reacting to each other:

$$\left\{ \begin{array}{lcl} x_{n+1} & = & 4a\,y_n(1 - y_n) \\ y_{n+1} & = & 4b\,x_n(1 - x_n) \end{array} \right.$$

for parameters $a, b \in [0, 1]$.

This is a coupled nonlinear system of difference equations.

looking for fixed points

$$\left\{ \begin{array}{rcl} x_{n+1} & = & 4ay_n(1-y_n) \\ y_{n+1} & = & 4bx_n(1-x_n) \end{array} \right.$$

Looking for fixed points, we eliminate y_n:

$$\begin{array}{rcl} x_{n+2} & = & 4\,a\,y_{n+1}(1-y_{n+1}) \\ & = & 16\,a\,b\,x_n(1-x_n)(1-4\,b\,x_n(1-x_n)) \end{array}$$

At the fixed point:

$$x = 16abx(1-x)(1-4\,bx(1-x)).$$

Dividing out the trivial solution $x = 0$ leads a cubic.

how many real roots?

An equation of degree three such as

$$1 = 16ab(1-x)(1-4bx(1-x))$$

has one or three real roots.

Exercise 4: Consider the following values for a and b.

1. $a = 0.8$, $b = 0.4$
2. $a = 0.8$, $b = 0.86$
3. $a = 0.8$, $b = 0.9$

- How many real roots in each case?
- For each case, make plots of x_n. Interpret the plots.

summary and bibliography

We started with a Jupyter notebook to model pursuit.

Then we considered linear and nonlinear systems.

We ended Chapter 9 of our text book.

- Charles R. MacCluer:
  Industrial Mathematics. Modeling in Industry, Science, and Government. Prentice Hall, 2000.